Competence without Performance: Representation Without Control on ARC-AGI-3

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Abstract

DreamerV3 world models master more than 150 control tasks under a single configuration, from Atari to collecting diamonds in Minecraft from scratch. Tiny Recursive Models of at most 7M parameters report 87% on Sudoku-Extreme, 85% on Maze-Hard and 45% on ARC-AGI-1, above language models orders of magnitude larger. We put both through one Gymnasium- and DreamerV3-compatible substrate for ARC-AGI-3, a benchmark of interactive games whose rules and goals must be inferred by playing them, and both collapse: 0.005 and 0.018 against a human-parity 1.0, over six games and two seeds, with cross-game pretraining giving no consistent benefit. We ask why, and separate the two answers a terminal score cannot tell apart. Each model has some of the abilities we measure, and neither has all of them. DreamerV3's reconstruction loss falls by at least 98% on five of six games and level identity is linearly decodable from its frozen latent above raw-pixel and elapsed-time controls on two games it never scores on, while its imagined rollouts fall below copying the last frame at every horizon on every game; the recursive model, trained on the same replays under a loss whose optimum is not the static scene, clones human play into non-zero scores and predicts transitions above that same baseline in 11 of 12 cells. The failure lies in reward discovery: on 21 of 25 games the reward is identically zero for the entire 500k-step run, imagined returns are constant, and the actor sits at its uniform maximum entropy of $\ln(4102)$ nats; score appears in exactly the four games where it departs. The two interventions that move anything, a factored action interface and a cloned human prior, act on that step rather than on the model: the factored interface ignites reward discovery on `cd82` in 6 of 6 seeds against 0 of 6 while making the world model's fit fifteen times worse. We read this as competence without performance. Two of its causes carry over to world models of physical scenes: a reconstruction loss that a mostly static scene satisfies by copying the observed frame, and a uniform-policy fixed point when reward arrives only on success.

1 Introduction

ARC-AGI-3 [1] is a benchmark of 25 public interactive abstract-reasoning games in which the rules and goals are *not* given and must be inferred through active play. We use two models in

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this work. DreamerV3 learns a latent world model and improves its policy inside it, and with one configuration and no per-task tuning it masters more than 150 tasks spanning Atari, DeepMind Control, Crafter, DMLab and Minecraft, where it is the first method to collect diamonds from scratch without human data [7]; on multi-step ARC-AGI-1 it beats PPO [9]. The Tiny Recursive Model takes the opposite approach, refining an answer by recursion at 5M to 7M parameters, and reports 87% on Sudoku-Extreme, 85% on Maze-Hard, 45% on ARC-AGI-1 and 8% on ARC-AGI-2, above language models orders of magnitude larger [8]. Between them they cover two common approaches: learn the dynamics and plan in them, or learn the solution and emit it.

We ran both models on ARC-AGI-3. It is scored by RHAE (Relative Human Action Efficiency): per level, $s_i = \min\big((\text{human_actions}/\text{ai_actions})^2,\ 1.15\big)$; a game score is the level-index-weighted mean $\sum_i s_i\, i / \sum_i i$ over all levels including uncompleted ones, and the total averages games equally, so 1.0 is human action efficiency. Both models land near the floor: 0.005 for DreamerV3 from scratch, 0.006 warm-started from a cross-game world model, 0.018 for the recursive model cloning human play, 0.007 for the same recursive model used as a learned simulator (Table 1). The ARC-AGI living survey [14] reviews $\sim$80 papers, documents a consistent two- to threefold performance drop from ARC-AGI-1 to ARC-AGI-2 across paradigms, finds only 3 that report any ARC-AGI-3 result, and describes ARC-AGI-3 as demanding exploration and world-model induction [16], which no current architecture demonstrates. Our two data points fall far below human efficiency, and not for want of capacity: these architectures do not fail at Sudoku, at Atari, or at ARC-AGI-1.

A final accuracy score is not enough, because it reports the same zero whether a model never built any account of the environment or built a usable one and could not act on it. Those are opposite diagnoses with opposite remedies, and psychology has a name for the confusion: *competence versus performance* [5, 10, 11].

An agent which cannot reason about physical objects and the consequences of its own actions cannot act reliably no matter how well it predicts the next frame. We instrument the two sides separately, on frozen checkpoints and against baselines that carry no world knowledge, and the answer is one-sided without being uniform: what these models represent transfers, what DreamerV3 simulates does not, and neither becomes control (§6 separates representational, dynamics and task competence). Its frames reconstruct and its latent carries level identity above raw-pixel and elapsed-time controls, its imagined rollouts lose to copying the last frame on every game, the recursive model clones human play into score and does beat that baseline, and none of it appears in RHAE. What does appear in RHAE is whether the actor ever leaves its uniform prior, which happens in exactly the 4 of 25 games that score at all.

(1) Substrate.[2] The first Gymnasium- and DreamerV3-`embodied`-compatible wrapper for ARC-AGI-3, which is what makes the two families comparable at all: a 4102-way flat discrete action space (parameter-less `ACTION1-5`; `ACTION6` over the 64×64 click grid; `ACTION7`/undo) with an optional per-game boolean mask, a $(64, 64, 3)$ `uint8` observation in DreamerV3's native modality, an offline loader for the 340-episode human replay corpus, and a post-hoc RHAE harness. Action coding, loader and metric are property- and regression-tested (318-test suite green). DreamerV3 is used stock and unforked at `size12m`, its smallest standard configuration.

(2) The first model-based RL result on ARC-AGI-3, paired against a supervised control: five arms over six games and two seeds, cold against cross-game pretrained, plus a 25-game single-seed sweep. Cross-game pretraining gives the RL arm no detectable benefit: it lifts none of the five zero-score games off the floor, and on the sixth its sign flips between the two seeds.

(3) A located competence–performance gap. Systematic measurements across both sides of the competence and performance divide, each verified against null controls, demonstrating that representational machinery transfers across games while the closed loop from acting to reward to commitment fails. We further test structural interventions that change reward discovery without improving the model.

2 Setup

Unified environment substrate and evaluation. To keep the arms comparable across paradigms, all of them act through a shared Gymnasium-compatible interface for ARC-AGI-3. Agents receive

[2] Code, replay loader and RHAE harness: `https://github.com/ssrhaso/arc3-wm`.

Table 1: The same two architectures, elsewhere and here. Left, results reported by their own authors on benchmarks outside ARC-AGI-3; right, what they score on ours, on one shared substrate and metric, where 1.0 is human action efficiency. The right column is our own per-game instantiation of each architecture, not a published checkpoint: 12M-parameter DreamerV3 at `size12m` and 7.5M/7.4M recursive models trained on the ARC-AGI-3 replay corpus.

Model	Reported elsewhere	Here (RHAE)
DreamerV3 [7]	150+ tasks under one configuration: Atari, DeepMind Control, Crafter, DMLab; first to collect Minecraft diamonds from scratch. Beats PPO on multi-step ARC-AGI-1 [9]	0.005 from scratch 0.006 pretrained
TRM [8]	87% Sudoku-Extreme, 85% Maze-Hard, 45% ARC-AGI-1, 8% ARC-AGI-2, above far larger language models	0.018 cloning play 0.007 as a simulator
Human		1.000
Random		0.000

$(64, 64, 3)$ `uint8` visual observations and select actions from a 4102-way flat discrete action space. Performance is evaluated with the post-hoc RHAE harness, where human baseline references correspond to the upper-median action counts of first-time human players (covering 70.5% of valid levels). We focus our core paired evaluation on six representative games (`vc33`, `sb26`, `cd82`, `tn36`, `ls20`, `lf52`) under the native, unshaped level-clear reward signal $r = \Delta\text{levels}$.

Experimental arms. We evaluate five configurations spanning model-based reinforcement learning and supervised imitation (Appendix Table 5), which vary in whether they learn an explicit transition model and in the policy objective. Three RL arms run DreamerV3 and train the policy in imagination on $r = \Delta\text{levels}$: *cold* from scratch, *warm* from the cross-game world model, and *factored*, with a (type, x, y) head in place of the flat 4102-way one. Two supervised arms, described below, never see reward, which makes them a control on exploration rather than a competitor.

DreamerV3 learns a Recurrent State-Space Model: an encoder $z_t \sim q_\phi(z_t \mid h_t, x_t)$ compressing the frame into 32×32 categorical latents, a recurrence $h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$, a dynamics prior $\hat{z}_t \sim p_\phi(\hat{z}_t \mid h_t)$ that must predict the latent *without* seeing the frame, and decoder, reward and continue heads on $s_t = (h_t, z_t)$. The actor is trained only on rollouts the dynamics head imagines forward, and the environment is never touched during policy improvement [7]. That decoupling is what makes the failure diagnosable at all: the reconstruction and dynamics losses report whether the model fits, RHAE reports whether the policy acts, and the two can be read separately. The warm arm initialises the world model from a single model pretrained self-supervised on the pooled 340-replay corpus of all 25 games and is otherwise identical to cold, making the warm start the only difference between the arms.

We use two separately trained Tiny Recursive Model (TRM)-based architectures [8] as supervised arms. The first, *TRM clone*, predicts the human action sequence as a 32-step plan, and provides a supervised control for how a strong recursive architecture performs on ARC-AGI-3. The second, *TRM-WM*, tests whether the same recursive architecture can be repurposed as a world-model architecture. Given the current palette grid and one flat action, it recursively refines a prediction of the next grid as per-cell 16-way colour logits. Used as a controller, it scores each candidate action by one forward step.

3 Performance

Baseline performance and policy entropy collapse. Across the evaluated benchmarks, DreamerV3 registers non-zero performance exclusively on `vc33`, where the marginal difference between warm and cold initializations displays conflicting signs across seeds (seed 0 favors warm by $+0.014$, while seed 1 favors cold by $+0.002$). Two seeds preclude formal variance estimation, so we read this as no detectable benefit rather than evidence of no effect; pretraining also lifts none of the other five games off the floor in any of their ten warm cells. Those zeros are structural rather than stochastic: on the five zero-RHAE games, the actor's policy entropy remains pinned at the uniform maximum ($\ln(4102) = 8.32$ nats) at 500k steps with a random-action fraction of exactly 1.000 (Fig. 2, right), indicating that these trajectories are indistinguishable from uniform random policies. Conversely,

Table 2: RHAE at 500k env-steps (RL arms) and over 30 live episodes per cell (supervised arms), mean of 2 seeds. A random agent that clears no level scores 0; an agent at human action efficiency scores 1.0 per level, with the 1.15 cap reserved for super-human efficiency. Every arm sits two orders of magnitude below parity, and every row maximum (bold) falls in a column with no transition model.

	DreamerV3 (RL)		TRM clone (supervised)		TRM-WM
Game	cold	warm	cold	warm	lookahead
vc33	0.0297	0.0357	**0.0540**	0.0366	0.0138
sb26	0.0000	0.0000	0.0000	0.0000	0.0000
cd82	0.0000	0.0000	0.0000	**0.0323**	0.0274
tn36	0.0000	0.0000	**0.0384**	0.0293	0.0000
ls20	0.0000	0.0000	0.0000	**0.0223**	0.0000
lf52	0.0000	0.0000	**0.0178**	0.0019	0.0000
Combined	0.0049	0.0059	0.0184	**0.0204**	0.0069
Random			0.0000		
Human			1.0000		

on vc33, where policy entropy decreases ($8.32 \rightarrow 6.3$, random-action fraction 0.75), a dedicated random-policy evaluation across $1{,}132$ episodes yields an RHAE of 0.0, confirming that its non-zero score is a genuine departure from a measured floor.

Correlation between policy entropy and task completion. We also ran the cold RL arm at seed 0 on all 25 public games. Four scored non-zero, all of them games whose engine offers the click grid: ft09 0.115, vc33 0.055, lp85 0.041 and r11l 0.018, each clearing exactly one level.[3] Normalised policy entropy is below 1.000 on precisely those four games and exactly 1.000 on the other 21, while reconstruction loss falls sharply on all 25, failures included. Score appears if and only if the actor departs its uniform prior, and nothing in the world-model loss predicts which games those will be.

Supervised imitation as an exploratory control. The supervised clone scores more than three times DreamerV3's cold-start score (0.0184 vs. 0.0049), with non-zero performance on tn36 and lf52, where every RL run scores zero. This makes it a diagnostic control. Because non-zero scores appear under the same environment, interface and metric once online exploration is removed, the environment itself is not the barrier, which points to online exploration as the point of failure; model and training method change together here, so this does not isolate a single cause.

4 Competence

If ARC-AGI-3 simply exceeded what these architectures can hold, they should fail on both sides of the divide, at representing the games as well as at playing them. They do not. We freeze the per-game seed-0 warm DreamerV3 checkpoint, and the per-game recursive models, and measure four things, each against a control that would pass without any world knowledge (Fig. 1). Three of the four transfer.

Convergence of visual state reconstruction. The training curves show that visual reconstruction loss collapses by at least 98% on five of the six games, from between 31 and 63 at the first logged window to between 0.06 and 0.74 at 500k steps, ranges taken across games (Fig. 2, left; per-game final values in Appendix Table 6). The dynamics KL settles between 1.0 and 1.6 on every game with terminal slopes below 0.013 per 1k steps, so the losses have plateaued rather than stopped mid-descent. Warm-started models begin 20 to 30 times lower than cold ones (from 30 to 65, against 600 to 1950), but both configurations reach the same floors, so the benefit of pretraining is transient. tn36 is the sole outlier, plateauing at ~ 6.8 (an 89% reduction). Notably, reconstruction fidelity exhibits no correlation with control performance: cd82 achieves tighter visual fit than vc33 (0.065 vs. 0.160)

[3] This sweep's cold arm pre-populates no replays; the paired study's cold arm does (Appendix A). The sweep's 0.055 on vc33 is therefore not comparable with the paired cold seed-0 cell (0.0411). The paired *warm* seed-0 cell, which Table 3 reports, is 0.0548 (Appendix Table 6).

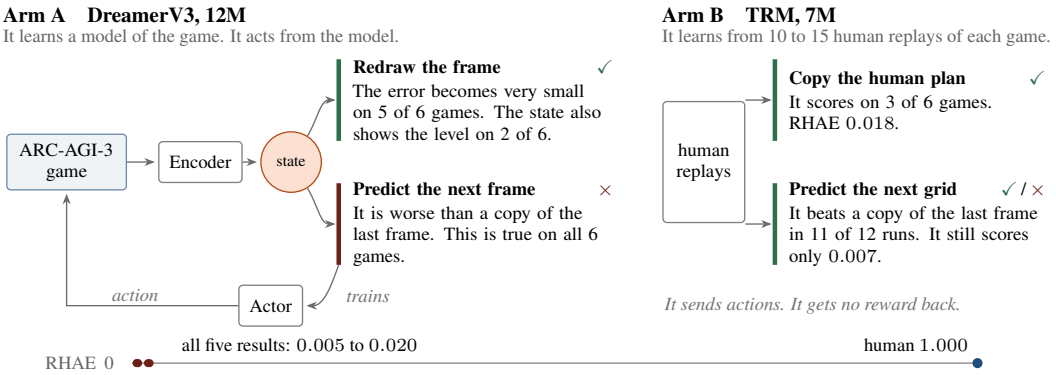


Figure 1: Two arms on one substrate. Arm A learns a model of the game and acts from it. Arm B copies human replays. A green bar shows a result better than a baseline that knows nothing about the game. A red bar shows a worse result. §4 gives the protocols.

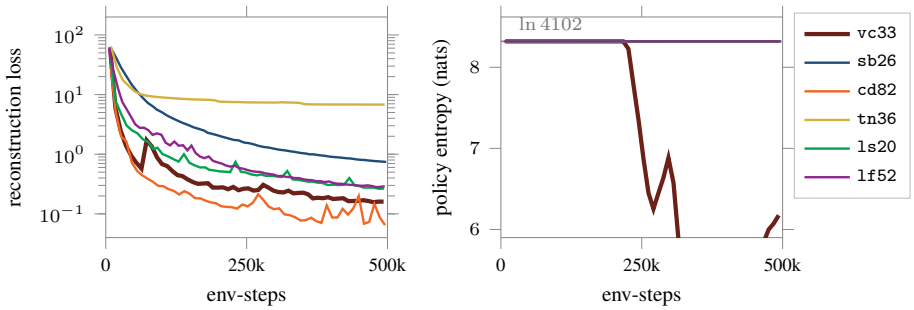


Figure 2: The two halves of the run, warm arm, seed 0. *Left:* reconstruction loss falls by two to three orders of magnitude on five of six games (`tn36` the lone plateau at $\sim$6.8). *Right:* policy entropy stays pinned at the uniform ceiling $\ln 4102 = 8.32$ nats for the entire budget on five of six games, whose traces are exactly superposed on the ceiling line; only `vc33` departs, and only to 6.3, still acting randomly on three-quarters of steps.

yet achieves zero RHAE, while `tn36` and `ls20` both fail entirely despite substantial differences in reconstruction loss.

Linear decodability of abstract task states. To evaluate internal state representations, we train cross-validated ridge classifiers on frozen latent features under 20-fold episode-grouped splits, defining a target as decodable only when its bootstrap confidence interval exceeds a label-permutation null control alongside two trivial baselines: a raw-pixel projection probe and an elapsed-step clock probe (Appendix C). Level identity clears the permutation control on `sb26` (0.83), `cd82` (0.43), and `tn36` (0.43). However, because `sb26`'s raw-pixel baseline reaches 0.77, its decodability largely reflects superficial frame features; only `cd82` and `tn36` decode significantly above both pixel and step controls, despite neither yielding non-zero policy performance. Furthermore, level-transition events are decodable on `cd82` (0.88), `sb26` (0.84), `vc33` (0.83), and `lf52` (0.63). The clock probe fails across all configurations, confirming that the latent state captures genuine environmental progression rather than trivial step counts.

Supervised learning from human replays. The per-game human replays that let the TRM clone score where no RL run does (§3) also support transition learning: trained with a cross-entropy loss that up-weights changed cells, TRM-WM outperforms copy-last-frame at $H{=}8$ in 11 of 12 game-seed cells on *held-out* episodes (combined 0.967 vs. 0.916; Fig. 3, right). These results are modest in absolute terms, but they establish the diagnostic point needed here: the replay corpus, interface, and evaluation metric support non-trivial learning. Zero performance elsewhere therefore cannot be attributed simply to the benchmark being unlearnable.

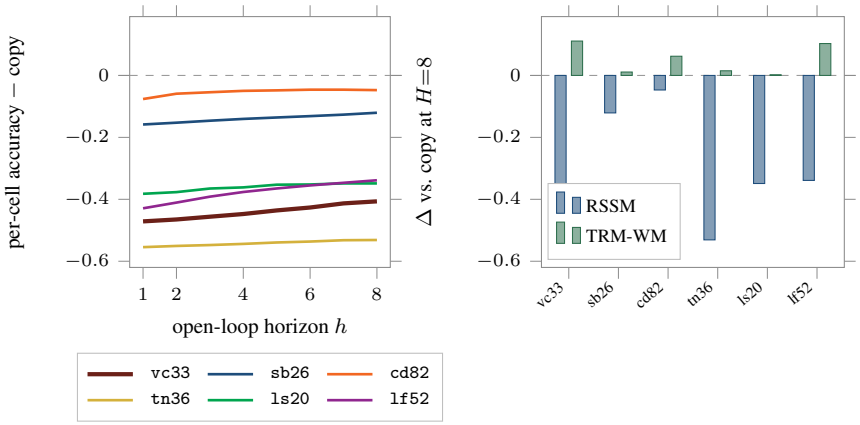


Figure 3: Open-loop transition fidelity relative to copy-last-frame; above the dashed line is better than copying. *Left:* DreamerV3's RSSM at horizons 1 to 8, one line per game. *Right:* the gap at H=8 for the RSSM and for TRM-WM (two-seed mean). Both use the same metric and baseline but different windows (TRM-WM on held-out episodes), so the signs of the gaps are comparable and the magnitudes are not.

Predictive failure of open-loop rollout simulation. We evaluate multi-step dynamics by encoding a four-frame context and imagining H=8 steps forward under ground-truth human action sequences without subsequent observations, measuring quantised per-cell accuracy against a copy-last-frame baseline on held-out episodes. Across all games and horizons, DreamerV3's open-loop rollouts underperform the copy baseline (Fig. 3, left), with accuracy deficits at H=8 ranging from -0.047 on `cd82` to -0.531 on `tn36`. This deficit does not grow with the horizon: the gap is maximal at h=1 and narrows slightly at longer horizons because the static copy baseline decays as the environment evolves, whereas the world model's prediction remains nearly static (`tn36`: $0.435 \rightarrow 0.423$ across eight steps; Fig. 4). This is not due to numerical model degeneracy; on random-policy rollouts where board states change by only $\sim 0.25\%$ of cells per step, both the model and copy baseline achieve ≈ 0.99 accuracy. Nor is it an artifact of copy being strong on static boards: on the two games with the most motion and the weakest copy baselines, `lf52` and `vc33`, the deficits are still 0.339 and 0.407 (Appendix Table 8). The predictive failure occurs when the board undergoes real state transitions, which is the regime model-based planning needs.

Action-conditioning and counterfactual discrimination. Planning requires an internal model capable of distinguishing between counterfactual actions. Evaluating $1{,}500$ real states per game, we probe two properties: *sensitivity* (the proportion of cells whose single-step prediction varies across candidate actions) and *discrimination* (whether the executed action predicts the subsequent state more accurately than alternative actions or the static copy baseline). Sensitivity reaches 0.084 on `cd82` and 0.063 on `vc33`, comparable to the 0.024 to 0.030 fraction of cells that change in ground-truth transitions. However, sensitivity drops to 0.005 on `sb26` and 0.0004 on `tn36`, indicating near-total action insensitivity. Counterfactual discrimination fails on every game probed: the prediction accuracy of the executed action differs from the mean of candidate alternatives by only 0.0004 to 0.008 (0.8867 vs. 0.8812 on `cd82`), outperforming copy on only 0.3% to 5.3% of states. Under random action policies, sensitivity is identically zero across all environments. Policy optimization therefore takes place inside an imagined environment that approximates an action-invariant static scene (counterfactual probes cover four of six games, excluding `ls20` and `lf52`).

Divergence between visual representation and causal dynamics. These findings explain why large drops in reconstruction loss do not yield working policies. Reconstruction loss is evaluated on the posterior encoding path, where the latent state directly conditions on the observed frame; on predominantly static boards, a decoder can minimize pixel loss simply by memorizing the current frame without modeling transition dynamics. In contrast, TRM trained with a change-weighted loss surpasses the copy-last-frame baseline on identical demonstrations, indicating that standard pixel losses are biased toward static solutions. However, better simulation does not translate into performance: deployed as a one-step lookahead controller, the recursive transition model achieves

Table 3: Everything we measured on the frozen models, per game. RHAE and the fit column are the seed-0 warm DreamerV3 checkpoint (Appendix Table 6); the rollout, action-conditioning and decode columns are measurements on that checkpoint, except TRM-WM, which is the two-seed mean of the discrete recursive model on held-out episodes. "sens." is the percentage of cells whose one-step prediction changes with the action (0 is action-blind), against 2.4–3.0% of cells that genuinely change in one step; "beats copy" is the share of states where the taken action's prediction beats copy-last-frame. † marks a decode above the label-permutation control, **bold** one that also exceeds both the raw-pixel and clock baselines. The "reward" column records whether reward proximity decodes from the latent above its own control (✓) or does not (×), from Table 11. Nothing in the table predicts the RHAE column.

		fit	rollout Δ vs. copy		action-conditioning		linear decode		
Game	RHAE	recon	RSSM	TRM-WM	sens.	beats copy	level id	best triv.	reward
vc33	**0.055**	0.16	−0.407	+0.111	6.3%	3.7%	0.37	0.25	×
sb26	0.000	0.74	−0.121	+0.011	0.5%	3.3%	0.83†	0.77	×
cd82	0.000	**0.06**	−0.047	+0.062	8.4%	5.3%	**0.43**†	0.37	×
tn36	0.000	6.80	−0.531	+0.015	0.04%	0.3%	**0.43**†	0.33	×
ls20	0.000	0.27	−0.349	+0.002	–	–	0.23	0.42	✓
lf52	0.000	0.29	−0.339	+0.103	–	–	0.35	0.31	✓

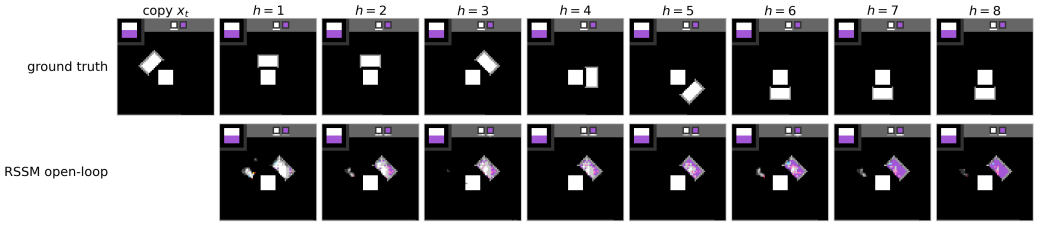


Figure 4: What the world model imagines on cd82. Top: ground truth $x_{t+1..t+8}$. Bottom: the RSSM's open-loop decode under the identical real action sequence from a four-frame context, with the copy-last-frame image x_t at the left. The imagined row is wrong from the first step and then stays nearly fixed while the true board moves. This row is the *prior* path; the loss curve of Fig. 2 measures the posterior path, on which this checkpoint has the study's lowest final loss (0.065, Table 3).

a combined RHAE of only 0.0069 (0.0048 when hybridized with the cloned prior), on par with DreamerV3 and well below the model-free imitation clone (0.0184). Thus, improved transition modeling alone is insufficient to resolve policy failure.

5 Diagnosis

Atari, DeepMind Control, Crafter, Minecraft, Sudoku-Extreme, Maze-Hard and ARC-AGI-1 differ in almost every respect, but each supplies one of two things that ARC-AGI-3 withholds, and each of our two families depends on one of them.

Extremely sparse reward landscapes. In the environments where DreamerV3 is strong, behaviour that is close to random already earns something: Atari scores tick over under arbitrary play, Crafter's early achievements fall out of random actions, control tasks pay continuously, and the Minecraft result is driven by a ladder of twelve intermediate milestones. ARC-AGI-3 pays only on a level clear. On 21 of the 25 games in the single-seed cold sweep, the reward received over the entire 500k-step run is identically zero, and the consequence is not slow learning but an equilibrium. DreamerV3's actor is improved only on imagined rollouts scored by the learned reward head; with no reward ever observed, that head predicts zero everywhere, imagined returns are constant, advantages vanish, and the only surviving gradient on the actor is its entropy regulariser, whose maximiser is the uniform policy. The logged entropy of exactly $\ln(4102) = 8.32$ nats and random-action fraction of exactly 1.000 are that fixed point. This matters for the obvious objection that 500k steps is simply too few: more steps of the same kind produce more zero-advantage updates, and nothing in the update rule

Table 4: Interventions on the same substrate, separated into *ignition* (the agent encounters reward at all during training) and *consolidation* (the behaviour survives into evaluation).

Intervention	Arm	Ignition	Consolidation (RHAE / seeds)
Cross-game WM pretrain	DreamerV3	unchanged	$0.0049 \to 0.0059$, sign flips by seed
Cross-game plan pretrain	TRM clone	n/a	$0.0184 \to 0.0204$, `ls20` and `cd82` leave the floor
Factored (type, x, y) head	DreamerV3	`cd82` $0/6 \to 6/6$, `sp80` $0/3 \to 3/3$	`cd82` $0/6 \to 4/6$, `sp80` $0/3$, `ft09` $1/3 \to 2/3$
Pixel-derived explore prior	DreamerV3	no gain over factored alone	below factored on 2 of 3 games
Factored / flat head	TRM clone	n/a	0.0184 / 0.0130 / 0.0071
Per-game action mask	TRM clone	n/a	`ls20` 0.0000, null
One-step lookahead	TRM-WM	n/a	0.0069 (0.0048 hybridised)

breaks the symmetry. What is missing is not compute but contact with reward, and a single brush with it is not enough: the two `sb26` cells that log one chance level clear apiece finish at exactly the same uniform entropy as the cells that log none (Appendix B). Consistently, normalised entropy is below the ceiling on precisely the four games of twenty-five that score at all, and at the ceiling on the twenty-one that do not.

Absence of explicit supervision targets. The recursive model's reported successes are all static input-output problems: a Sudoku grid, a maze, each with a ground-truth answer to refine towards. ARC-AGI-3 supplies no target at all unless human play is treated as one, and the corpus holds roughly ten to fifteen demonstrations per game. That thin supply is nonetheless enough to make the clone the best arm in this paper. Supplying the missing target is what moves the number; supplying a better simulator, as §4 showed, is not.

High-dimensional unlabelled action spaces. Their semantics are unstated and game-dependent, against 18 actions in Atari, 17 in Crafter and a handful of continuous dimensions in control. Five indices are parameter-less actions, $4096 = 64 \times 64$ separately encode every screen coordinate for `ACTION6`, and one is undo, so "click at $(20, 30)$" and "click at $(20, 31)$" are statistically unrelated labels. On `ls20` only 4 of the 4102 change the board at all.

Structural action factorisation. Two interventions move something, and neither improves the model's fit (Table 4). Replacing the flat 4102-way categorical with the stationary factorisation $a_t = (\text{type}_t, x_t, y_t)$, where the type head ranges over the game's action types and two coordinate heads range over 64 positions each, keeps every action representable and changes only how an action is expressed, never which action solves the game. On `cd82` it changes reward discovery outright: across a balanced six-versus-six comparison all $6/6$ factored agents clear level 1 at least once during training and $0/6$ flat agents ever do, factored agents develop positive imagined return (median maximum 1.6399) and leave their entropy floor while every flat actor's imagined return is exactly zero, and $4/6$ factored seeds against $0/6$ flat carry a clear into evaluation (median RHAE 0.0548 vs. 0.0000, directional Fisher exact $p = 0.0303$; `cd82` was selected after a promising $2/2$ against $0/2$ pilot, and on the four genuinely new seeds alone it is $2/4$ against $0/4$, $p = 0.2143$). The decisive detail is that the model gets *worse* while the behaviour appears: median final image loss is 2.8213 for factored against 0.1805 for flat. The second is the cloned prior: warm-starting the clone from a pretrain pooled over all 25 games lifts `ls20` (both seeds) and `cd82` (one) off the floor, at some cost to `lf52` (Table 2). That answers the RL arm's pretraining null: the pooled corpus does contain cross-game structure worth transferring, and the RL controller could not cash it in.

Decoupling reward ignition from policy consolidation. These runs separate *ignition*, encountering reward, learning a non-zero reward model, assigning positive return to some imagined future and moving off the uniform prior, from *consolidation*, retaining that behaviour and expressing it under held-out evaluation. Factorisation reliably changes the first and unreliably the second. A stratified follow-up, frozen before its runs, took one button-only game (`g50t`), one click-only (`ft09`) and one mixed (`sp80`) at three seeds per cell: on `sp80` the flat arm never clears a level at all while the factored arm produces 16, 16 and 32 positive training episodes and still scores zero at evaluation in all three seeds; `ft09` improves from $1/3$ to $2/3$ successful seeds; `g50t` never clears under either arm, showing that a small exposed button grammar does not by itself solve hard exploration. The

pre-registered endpoint gate fails, as does one for a training-only pixel-derived exploration prior added on top (Appendix E). An agent that never found reward and an agent that found it and lost it both report RHAE 0, and they are different bugs with different fixes. The flat encoding therefore explains why reward is never found on `cd82` and `sp80`, but not the rest of the failure: factored agents lose what they find on `sp80`, `g50t` stays at zero under both heads, and the factored median on `cd82` (0.0548) is $\sim 5\%$ of human efficiency.

Action-space masking and exploration priors. On `ls20` only the four directional actions change the board and the 4096 click cells are never offered (Appendix F), so a uniform policy changes the board about once in $1{,}025$ steps. The obvious fix is the per-game mask our substrate already exposes, and a 2×2 test in the supervised arm resolves it as a null: masked training and evaluation alone leaves `ls20` at 0.0000 on both seeds, cross-game pretraining lifts it to clears on both seeds under the *unmasked* interface, and adding the mask on top of pretraining does not improve on the unmasked pair. What `ls20` was missing is a prior over its four live actions, not a smaller action space.

Taxonomy of failure modes across benchmark games. The six games do not all fail for the same reason, and a frozen-latent probe partitions them independently of the training logs. We asked whether a level clear falls within the next five steps, and within a wider ten-step window, using the same linear protocol and permutation control as above. It is decodable on `ls20` and `lf52` alone, and not on `cd82`, `sb26`, `tn36` or `vc33`. The four where reward proximity is undecodable are exactly the four the training logs diagnose as exploration-bound (three never leave uniform and `vc33` barely does); the two where it is decodable are exactly the two this section diagnoses separately, a diluted action space and, on `lf52`, a hard 64-action level-1 budget that terminates all 105 evaluation episodes at exactly the same step (Appendix F). Two independent lines of evidence draw the same line through the same six games, which is what makes exploration the licensed next intervention on four of them and the wrong one on the other two.

6 Discussion

Competence and performance. Developmental psychology has a long record of reading a missing behaviour as a missing capacity, and of later tracing that behaviour to performance factors rather than to a missing concept [5, 10]. We state the capacity operationally: a model is competent for a quantity at a horizon under a decoder class if that quantity is recoverable from its state by such a decoder, above a named baseline that carries no world knowledge. The definition separates three competences that the word usually runs together. *Representational* competence, recovering the current state, DreamerV3 has in part: its frames reconstruct, and level identity decodes above pixel and clock baselines on two of six games. *Dynamics* competence, predicting what an action will do, DreamerV3 lacks, since its rollouts lose to copy-last-frame on every game and fail to discriminate actions on every game probed, while TRM-WM has it in the narrow sense of beating copy in 11 of 12 cells. *Task* competence, which RHAE measures, neither has: the best arm reaches $\sim 2\%$ of human efficiency. No model here holds all three; our claim is that every competence we can measure, in either model, fails to become performance, not that either model understands these games. This is a claim about two named architectures at one budget, not about world models or model-based RL in general: 500k steps and $N = 6 \times 2$, plus 25 games at one seed, scope it to stock DreamerV3 with and without this pretraining and to 7M-parameter recursive models trained on this corpus. The prior results in Table 1 are context rather than controlled comparison; our probes are linear, the counterfactual measurement one-step over four of six games, and the recursive model a depth-one lookahead controller, so it bounds shallow planning only.

What carries over to physical environments. ARC-AGI-3's grids are abstract, but two findings do not depend on them. First, a posterior-path reconstruction loss can fall by orders of magnitude while open-loop prediction stays below copy-last-frame, because a decoder can reproduce a mostly static frame it was given. Physical video such as CLEVRER [15] and Physion [3], where most pixels do not change between frames, shares this property, so scoring open-loop rollouts against copy-last-frame is a cheap check for any pixel-reconstruction world model [6]. Second, an actor trained in imagination under success-only reward sits at the uniform policy until a success occurs, in physical tasks such as PHYRE [2] as much as in level clears, and only denser reward or reward-free exploration breaks that fixed point.

The measurements say which remedy applies where: on the four exploration-bound games, whose latent does not carry reward proximity, reward-free exploration, for which Plan2Explore [12] is the Dreamer-family baseline; on `ls20` and `lf52` a prior over live actions and the level-1 action budget instead; and on the model side, losses whose optimum is not the static scene, together with core-knowledge priors [4, 13]. We have not run the two direct tests: Plan2Explore, and DreamerV3 trained with TRM-WM's change-weighted loss. Across six games, five arms and two architectures, better prediction did not once become better action.

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A Protocol

Table 5: The five arms. Only the RL arms consume reward; the supervised arms never see one, which is what makes them a control on exploration rather than a competitor.

Arm	Transition model	Policy trained by	Action head	Params
DreamerV3 cold	RSSM	imagination, $r = \Delta$levels	flat 4102	12M
DreamerV3 warm	RSSM, 25-game pretrain	imagination, $r = \Delta$levels	flat 4102	12M
DreamerV3 factored	RSSM	imagination, $r = \Delta$levels	(type, x, y)	12M
TRM clone	none	supervised on human replays	spatial map	7.5M
TRM-WM	discrete TRM	one-step lookahead, no learning	spatial map	7.4M

Reinforcement-learning arms. Stock, unforked DreamerV3 at the `size12m` configuration, registered as an `embodied` environment through our wrapper, 500k environment steps per run at default hyperparameters, native level-clear reward, two seeds per cell. The warm arm loads a world model pretrained self-supervised (encoder, RSSM, decoders, reward and continue heads only; actor and critic not updated) on the pooled 340-replay buffer of all 25 public games, on one A100, with image-reconstruction loss settling at 53.66 on the mixed buffer. Each per-game run then initialises from that checkpoint, initialises a fresh actor and critic, pre-populates the buffer with that game's human replays and begins online interaction; the cold arm is identical but for the warm start, which is the only difference between the arms. Both train and evaluation use the full unmasked 4102-way interface. RHAE is computed post-hoc from the per-episode reward streams captured by an evaluation-side wrapper, never in the training loop.

Supervised arms. *TRM clone*: a 7.5M-parameter Tiny Recursive Model [8] trained per game to predict the human action sequence as a 32-step plan, under an episode-grouped 90/10 train/validation split with early stopping. Evaluation decodes the executed plan slot greedily over 30 live episodes per cell, under the same unmasked interface and the same post-hoc RHAE harness as the RL arms, so no replay data is touched at scoring time. The warm variant first pretrains an identical model on the pooled replay corpus of all 25 public games and then finetunes per game exactly as above; the seeded per-game split is identical in both stages, so the finetune's early-stopping episodes are never trained on. *TRM-WM*: a 7.4M-parameter model that, given the current palette grid and one flat action, recursively refines a prediction of the next grid as per-cell 16-way colour logits, with auxiliary level-clear, terminal-state and per-cell-change heads and a cross-entropy that up-weights cells changing between frames by $20\times$. It uses the same replays and the same episode-grouped split; rollout fidelity is probed on held-out episodes only.

Metric. RHAE human baselines are the upper-median first-time-player action count per level, taken from the human replay corpus. Levels with fewer than two completers are dropped from both the numerator and the denominator as statistically indefensible upper-medians, giving coverage of $129/183$ levels (70.5%); every game retains at least two covered levels.

B Full reinforcement-learning sweep

The 25 public games divide by engine affordance into six button-only, six click-only and thirteen mixed-control games. All four scorers are click games, which points at the spatial action parameterisation as an intervention target (§5), though a single-seed sweep cannot establish a causal class effect.

C Probe details and full results

All probes run on frozen per-game checkpoints (warm, seed 0). The generative probes (P2, P3) encode a four-frame context and predict forward with no further observations; predictions are decoded, quantised to the game palette and scored as per-cell accuracy. Two sources are collected per game where environment files were cached: *human*, the held-out replay corpus, which carries goal-directed and board-changing dynamics and is the strong test, and *random*, fresh masked-uniform

Table 6: All 24 paired runs (6 games $\times$ {cold, warm} $\times$ 2 seeds), final-step values. Training episode counts range 502–9,839 and evaluation counts 16–28 per run, so raw clear counts are not comparable across rows; RHAE is the normalised quantity. **Bold** marks the four cells that deviate from the otherwise uniform off-vc33 pattern (see notes).

Game	Arm	Seed	Recon. (final)	Dyn. KL (final)	Policy ent. (nats)	Rand.-act. fraction	Train clears	Eval clears	RHAE
vc33	cold	0	0.661	1.290	6.481	0.779	33	2	0.0411
vc33	cold	1	0.178	1.119	6.372	0.766	17	3	0.0182
vc33	warm	0	0.160	1.244	6.276	0.754	17	4	**0.0548**
vc33	warm	1	0.135	1.143	6.517	0.783	17	2	0.0166
sb26	cold	0	0.992	1.139	8.319	1.000	0	0	0.0000
sb26	cold	1	3.058	1.635	**8.282**	**0.995**	**1**	0	0.0000
sb26	warm	0	0.742	1.549	8.319	1.000	0	0	0.0000
sb26	warm	1	1.904	2.375	8.319	1.000	**1**	0	0.0000
cd82	cold	0	0.183	1.001	8.319	1.000	0	0	0.0000
cd82	cold	1	0.185	1.002	8.319	1.000	0	0	0.0000
cd82	warm	0	**0.065**	1.033	8.319	1.000	0	0	0.0000
cd82	warm	1	0.062	1.162	8.319	1.000	0	0	0.0000
tn36	cold	0	1.052	1.001	8.319	1.000	0	0	0.0000
tn36	cold	1	1.003	1.001	8.319	1.000	0	0	0.0000
tn36	warm	0	**6.797**	1.014	8.319	1.000	0	0	0.0000
tn36	warm	1	**7.280**	1.014	8.319	1.000	0	0	0.0000
ls20	cold	0	0.388	1.139	8.319	1.000	0	0	0.0000
ls20	cold	1	0.338	1.007	8.319	1.000	0	0	0.0000
ls20	warm	0	0.266	1.006	8.319	1.000	0	0	0.0000
ls20	warm	1	0.295	1.049	8.319	1.000	0	0	0.0000
lf52	cold	0	0.309	1.001	8.319	1.000	0	0	0.0000
lf52	cold	1	0.360	1.001	8.319	1.000	0	0	0.0000
lf52	warm	0	0.290	1.006	8.319	1.000	0	0	0.0000
lf52	warm	1	0.274	1.007	8.319	1.000	0	0	0.0000

Notes. A random-action fraction of 1.000 means the policy is indistinguishable from uniform (entropy at the ceiling $\ln 4102 = 8.319$). The four bold deviations: sb26 cold-s1 and warm-s1 each log one training clear under a still-uniform policy and yield no evaluation clear (chance, not commitment); tn36's warm world model alone fails to reach the sub-1 reconstruction floor. Evaluation clears and RHAE have no exceptions: only vc33 is non-zero.

Table 7: Non-zero cells in the 25-game cold baseline (seed 0, 500k steps, no warm start and no replay pre-population). The remaining 21 games score exactly zero and clear no level. Normalised entropy is the actor's entropy relative to its uniform ceiling, so 1.000 is exactly uniform.

Game	Control class	RHAE	Levels	Norm. entropy
ft09	click	0.115	1	0.901
vc33	click	0.055	1	0.771
lp85	click	0.041	1	0.920
r11l	click	0.018	1	0.998
Other 21 games	button / click / mixed	0	0	1.000

rollouts, where boards change on $\sim 0.25\%$ of cells per step and copy-last-frame is therefore an almost unbeatable baseline. The latent probes (P4) use ridge classifiers on standardised features with an episode-grouped 20-fold split, over the deterministic state, the stochastic state and their concatenation; a target counts as decodable only when the probe's bootstrap CI lower bound exceeds the upper bound of a label-permutation control run through the identical pipeline.

Level-transition events decode above the permutation control in cd82 (0.88), sb26 (0.84), vc33 (0.83) and lf52 (0.63), and not in tn36 or ls20.

vc33 is the sharpest of the four failures: its raw scores (0.638, 0.677) are the second-highest in the table, behind only lf52, and it still fails the bar, because the result is too variable across folds and its confidence interval overlaps the control's. That is informative rather than merely negative. The one

Table 8: P2, open-loop rollout fidelity. "changed" is the fraction of cells that genuinely differ from the context frame at $H{=}8$, i.e. how much work the copy baseline is doing. The model is below copy at both horizons on every game and both sources. On the random source, where the board is nearly static, model and copy are both ≈ 0.99, so the model is not degenerate; the deficit appears exactly where the board moves.

			$h{=}1$		$H{=}8$		
Game	Source	windows	model	copy	model	copy	changed
cd82	human	231	0.8935	0.9697	0.8770	0.9244	0.076
sb26	human	301	0.8284	0.9868	0.8216	0.9422	0.058
ls20	human	721	0.5910	0.9732	0.5797	0.9285	0.072
lf52	human	743	0.5303	0.9594	0.5254	0.8640	0.136
vc33	human	534	0.5044	0.9758	0.4676	0.8742	0.126
tn36	human	445	0.4346	0.9891	0.4231	0.9545	0.046
cd82	random	360	0.9894	0.9998	0.9897	0.9974	0.003
sb26	random	491	0.9864	0.9996	0.9860	0.9970	0.003
lf52	random	210	0.9928	0.9995	0.9928	0.9968	0.003
vc33	random	40	0.9940	0.9997	0.9945	0.9975	0.003

Table 9: P3, counterfactual action-sensitivity and discrimination, $1{,}500$ states per game on the human source. "sens." is the fraction of cells whose one-step prediction differs across candidate actions (0 is an action-blind copier). "taken" and "others" are the cell accuracies of the prediction under the action actually taken and the mean over the alternatives; a competent dynamics head should separate them and beat copy. It does neither. Under a random action source, sensitivity is exactly zero in every game. Counterfactual sets were not collected for ls20 and lf52.

Game	sens. (human)	taken	others	copy	taken beats copy	sens. (random)
cd82	0.0844	0.8867	0.8812	0.9676	5.3%	0.0000
vc33	0.0626	0.4904	0.4824	0.9759	3.7%	0.0000
sb26	0.0046	0.8125	0.8121	0.9867	3.3%	0.0000
tn36	0.0004	0.4965	0.4963	0.9928	0.3%	–

game in this paper whose controller escapes uniform behaviour shows no evidence that its escape is explained by decodable reward proximity, which argues against reward-adjacent latent structure being what lets vc33 commit at all.

D Supervised and discrete-world-model arms: full grids

These runs are protocol-matched to the RL arms in interface and metric: unmasked 4102-way action space, post-hoc RHAE, 30 live evaluation episodes per cell, and all values recomputed from raw per-episode logs.

Structured actions could help for two unrelated reasons: they make exploration geometrically easier, and they make the output statistically easier to learn. The supervised arm isolates the second, having neither exploration nor a transition model, and there combined RHAE degrades monotonically as output structure is removed (0.0184 spatial map, 0.0130 coordinate-factored, 0.0071 monolithic 4102-way), with the loss coverage-shaped: of the three games the spatial map scores on, the factored head retains only vc33 and the monolithic head vc33 plus a trace of tn36. Part of the effect is therefore statistical rather than exploratory. But the two channels land on disjoint games: under this cold cloning protocol every head scores zero on cd82 and sp80, the two games where the factored actor ignites in RL, so the ignition above is not the channel isolated here.

A step-0-weight variant of the warm arm clears cd82 on both seeds (0.1057, 0.0851); we report the protocol-matched arm above. Only sb26 is zero in every supervised cell, cold or warm, marking a failure mode that neither exploration nor demonstrations explain at this scale.

Table 10: P4, level-identity decoding against two trivial baselines. Chance is $1/(\#\text{levels})$. † marks a decode above the label-permutation control by CI separation. Level structure is decodable from the latent and absent from both the raw frame and the clock only in `cd82` and `tn36`; in `sb26` the raw-pixel baseline decodes nearly as well, so its signal is largely visible structure. The clock never passes control, so none of these are step counters.

Game	RHAE	chance	latent	pixel	clock
`cd82`	0.000	0.17	$\mathbf{0.43}^{\dagger}$	0.37	0.33
`tn36`	0.000	0.14	$\mathbf{0.43}^{\dagger}$	0.25	0.33
`sb26`	0.000	0.12	$0.83^{\dagger}$	$0.77^{\dagger}$	0.32
`vc33`	**0.055**	0.14	0.37	0.24	0.25
`lf52`	0.000	0.10	0.35	0.27	0.31
`ls20`	0.000	0.14	0.23	0.28	0.42

Table 11: Reward-proximity decodability (`deter+stoch` features, human replays, warm seed 0, 20-fold CV). `reward_imminence` asks whether a level clear falls within the next five steps; `reward_proximity` asks the same over $K{=}10$. Score is the mean balanced accuracy across folds; the control ceiling is the 97.5th-percentile balanced accuracy attainable by chance under this exact protocol and class balance, not a naive 0.5. † clears the control.

		`reward_imminence`		`reward_proximity`	
Game	Failure mode	score	control	score	control
`cd82`	exploration	0.614	0.576	0.606	0.582
`sb26`	exploration	0.622	0.631	0.613	0.581
`tn36`	exploration	0.517	0.574	0.570	0.563
`vc33`	exploration	0.638	0.559	0.677	0.616
`ls20`	action dilution	$0.520^{\dagger}$	0.503	$0.613^{\dagger}$	0.520
`lf52`	episode budget	$0.672^{\dagger}$	0.558	$0.709^{\dagger}$	0.537

E Action factorisation: full results

On the balanced `cd82` follow-up, flat A0 succeeds in $0/6$ seeds and factored B1 in $4/6$, with median RHAE 0 against 0.0548 and a directional Fisher exact test on all twelve runs giving $p = 0.0303$. This is not an independent confirmation, since `cd82` was selected after seeds 0 and 1 produced a $2/2$ against $0/2$ contrast; on the four genuinely new seeds alone B1 is $2/4$ against $0/4$ ($p = 0.2143$, one-sided). We therefore treat it as a strong pooled single-game signal whose post-selection holdout is inconclusive. The internal diagnostics are what localise the change: median final image loss is lower for A0 than B1 (0.1805 against 2.8213), A0's imagined maximum return remains exactly zero with every actor at its uniform floor, while all six B1 agents develop positive imagined return (median maximum 1.6399), depart the matched factored-uniform entropy floor, and clear level 1 at least once during training, including the two seeds that finish with zero evaluation clears. On `ft09`, B1's median is 0.00304 against A0's zero (the non-zero B1 scores are 0.115 and 0.00304).

F Failure modes on the two static-board games

Table 12: Paired RHAE, supervised TRM plan-cloning arm. `ls20` s0 comes from the pilot run under the identical recipe.

	Cold		Warm			
Game	s0	s1	s0	s1	Cold $\bar{x}$	Warm $\bar{x}$
`vc33`	0.0314	**0.0766**	0.0579	0.0152	**0.0540**	0.0366
`sb26`	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
`cd82`	0.0000	0.0000	0.0645	0.0000	0.0000	0.0323
`tn36`	0.0357	0.0411	0.0229	0.0357	0.0384	0.0293
`ls20`	0.0000	0.0000	0.0035	0.0411	0.0000	0.0223
`lf52`	0.0133	0.0222	0.0018	0.0021	0.0178	0.0019
Combined					**0.0184**	**0.0204**

Table 13: Open-loop rollout fidelity of the discrete TRM world model at $H{=}8$, per-cell accuracy on held-out episodes, against copy-last-frame on the same windows. The TRM model exceeds copy in 11 of 12 cells; the RSSM exceeds it in none (Table 8). Windows differ between the two probes, so the values are not directly comparable across tables.

	TRM-WM		Copy			
Game	s0	s1	s0	s1	TRM $\bar{x}$	Copy $\bar{x}$
`vc33`	0.9488	0.9866	0.8198	0.8936	0.9677	0.8567
`sb26`	0.9882	0.9711	0.9744	0.9633	0.9797	0.9689
`cd82`	0.9940	0.9733	0.9232	0.9199	0.9837	0.9216
`tn36`	0.9703	0.9358	0.9539	0.9229	0.9531	0.9384
`ls20`	0.9555	0.9648	0.9627	0.9528	0.9602	0.9578
`lf52`	0.9625	0.9501	0.8826	0.8247	0.9563	0.8537
Combined					**0.9668**	**0.9162**

Table 14: RHAE of the world-model-based controllers under the cold protocol of Table 12. *wm* scores each candidate action by one forward step of the frozen model (change-mask, level-clear and next-frame heads); *hybrid* rescales the cloned policy's action probabilities by the same one-step scores. Neither approaches the model-free plan-cloning arm (0.0184), and hybrid rescoring degrades the cloned prior (`vc33` s1: $0.0766 \rightarrow 0.0005$).

	wm		hybrid			
Game	s0	s1	s0	s1	wm $\bar{x}$	hybrid $\bar{x}$
`vc33`	0.0166	0.0110	0.0548	0.0005	0.0138	0.0277
`sb26`	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
`cd82`	0.0000	0.0548	0.0000	0.0000	0.0274	0.0000
`tn36`	0.0000	0.0000	0.0006	0.0000	0.0000	0.0003
`ls20`	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
`lf52`	0.0000	0.0000	0.0008	0.0012	0.0000	0.0010
Combined					**0.0069**	**0.0048**

Table 15: Paired open-loop fidelity at $H{=}8$: the per-game TRM world model trained cold against one warm-started from a 25-game pretrain under the identical recipe, probed on identical held-out windows within each seed. Warm-starting does not improve fidelity. † marks warm cells whose finetune early-stopped with frame exact-match pinned at 0, leaving a near-pretrain checkpoint; excluding them the mean warm minus cold difference is ≈ 0.000.

	Cold		Warm				
Game	s0	s1	s0	s1	Cold $\bar{x}$	Warm $\bar{x}$	Copy $\bar{x}$
vc33	0.9488	0.9866	0.9339	0.8793†	0.9677	0.9066	0.8567
sb26	0.9882	0.9711	0.9865	0.9153†	0.9797	0.9509	0.9689
cd82	0.9940	0.9733	0.9117†	0.9716	0.9837	0.9417	0.9216
tn36	0.9703	0.9358	0.9603	0.9566	0.9531	0.9585	0.9384
ls20	0.9555	0.9648	0.9518	0.9840	0.9602	0.9679	0.9578
lf52	0.9625	0.9501	0.9416	0.9666	0.9563	0.9541	0.8537
Combined					**0.9668**	**0.9466**	**0.9162**

Table 16: Supervised head ablation under the cold protocol of Table 12, two seeds per cell, median in parentheses. The spatial-map column is the Table 12 cold arm itself; sp80 is excluded from the combined score.

Game	Flat (A0 grammar)	Factored (B1 grammar)	Spatial map
vc33	.0269, .0548 (.0408)	.0315, .1247 (.0781)	.0314, .0766 (.0540)
sb26	0, 0 (0)	0, 0 (0)	0, 0 (0)
cd82	0, 0 (0)	0, 0 (0)	0, 0 (0)
tn36	.0016, .0016 (.0016)	0, 0 (0)	.0357, .0411 (.0384)
ls20	0, 0 (0)	0, 0 (0)	0, 0 (0)
lf52	0, 0 (0)	0, 0 (0)	.0133, .0222 (.0178)
Combined	.0071	.0130	.0184
sp80	0, 0 (0)	0, 0 (0)	0, 0 (0)

Table 17: Pre-registered controller pilot, two seeds and 500k steps per cell; parentheses show the median. A0 is the flat 4102-way baseline, B1 the factored (type, x, y) head, B2 the factored head plus a training-only pixel-derived exploration prior. The primary B2 gate (exceed A0 in median on at least two of three games, with a strict majority of B2 seeds clearing a level in every qualifying game) fails $1/3$.

Game	A0 flat	B1 factored	B2 factored + prior
vc33	.0411, .0182 (.0297)	.0248, .0548 (.0398)	.0140, .0548 (.0344)
cd82	0, 0 (0)	.0548, .0548 (.0548)	.0548, 0 (.0274)
sb26	0, 0 (0)	0, 0 (0)	0, 0 (0)

Table 18: Stratified A0 against B1 generalisation, frozen before its runs. "Train" and "eval" count seeds with at least one level clear. The endpoint gate fails because B1 has a higher evaluation success count only on ft09; the ignition effect nonetheless generalises to sp80.

		A0 flat		B1 factored	
Game	Class	Train	Eval	Train	Eval
g50t	button	0/3	0/3	0/3	0/3
ft09	click	3/3	1/3	3/3	2/3
sp80	mixed	0/3	0/3	3/3	0/3

(a) `ls20`: only 4 of 4102 action indices change the board

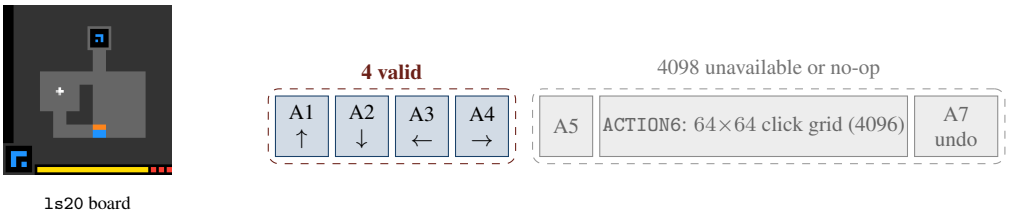


Figure 5: `ls20`'s engine offers only the four directional actions. `ACTION5` is a parameter-less no-op, `ACTION7` is undo, and the 4096 click cells are never offered at all, appearing in none of the 7,605 logged `ls20` frames, so they are masked-unavailable rather than merely inert. A uniform policy therefore changes the board on ≈0.1% of steps, about once in 1,025. §5 shows that supplying the mask does not fix the game and supplying a prior does.

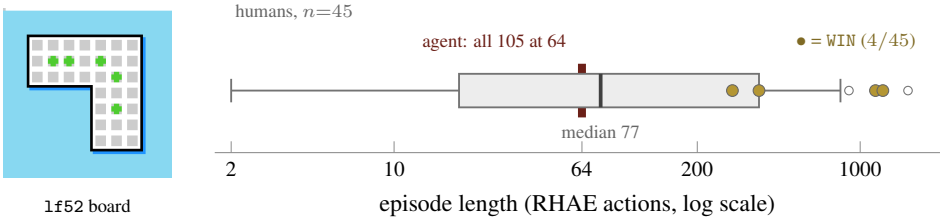
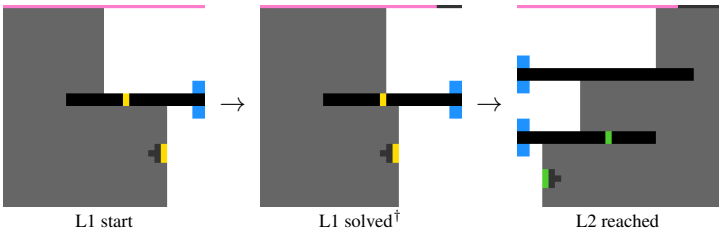


Figure 6: `lf52` episode lengths in RHAE action units. Every agent evaluation episode terminates at exactly 64 actions with zero variance (105/105); training episodes floor at 65, a reset-frame counting convention, with a thin tail to ≈236 and no level completions, and none approach the 1000-step truncation budget. These are terminations, not truncations: the engine enforces a hard per-level action budget (level 1 = 64; later tiers 64×5 and 64×10) and sets `GAME_OVER` when a non-clearing policy exhausts it. The budget is action-type weighted (directional and click moves cost 1, undo 20, no-ops 0), which desynchronises env-steps from budget count and produces the smooth tail. Human episodes (n=45 across 11 session replays) take a comparable median of 77 actions but a far wider spread (max 1604), reach `WIN` four times, and never trigger `GAME_OVER`. The contrast is therefore not episode length but determinism and outcome: under a uniform policy the agent exhausts the same fixed budget at the same step every episode, gaining nothing from repetition.

(a) `vc33`, the only game whose policy clears a level (RHAE 0.055)



(b) the policy keeps acting on the board and never clears a level (RHAE 0)

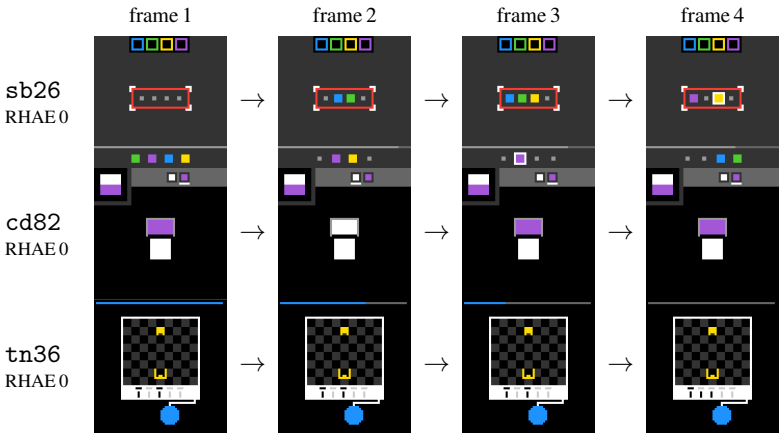


Figure 7: What the trained policies do. **(a)** `vc33` is the only game whose policy ever clears a level, starting level 1 and reaching level 2. †The solved frame is from a human replay; the other two are real trained-model observations. **(b)** On `sb26`, `cd82` and `tn36` the policy visibly acts on the board across one trained-model episode (seed-0 warm, four evenly spaced frames) and never reaches a goal. `ls20` and `lf52`, not shown, barely change the board at all for the separate reasons dissected in Figs. 5 and 6.